Mr Dimelow

3 Processes, techniques and specialist tools

3.4 Permanent and Semi-Permanent Joining Techniques

c) heat – oxy-acetylene welding, MIG welding, brazing, hard soldering, soft soldering (including use of specialist tools)

Gas-Shielded Welding







In this presentation you will:

- explore the welding processes that use gas shielding
- investigate the gases used in gas-shielded welding

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Introduction

Metal Inert Gas (MIG) welding is a versatile process, typically used in construction, shipbuilding and automotive industries.

Tungsten Inert Gas (TIG) welding is a specialist process, typically used for welding thin steels and alloys, and metals such as gold, chrome, cobalt and titanium.

Both processes use gases to shield welds from air contaminants.





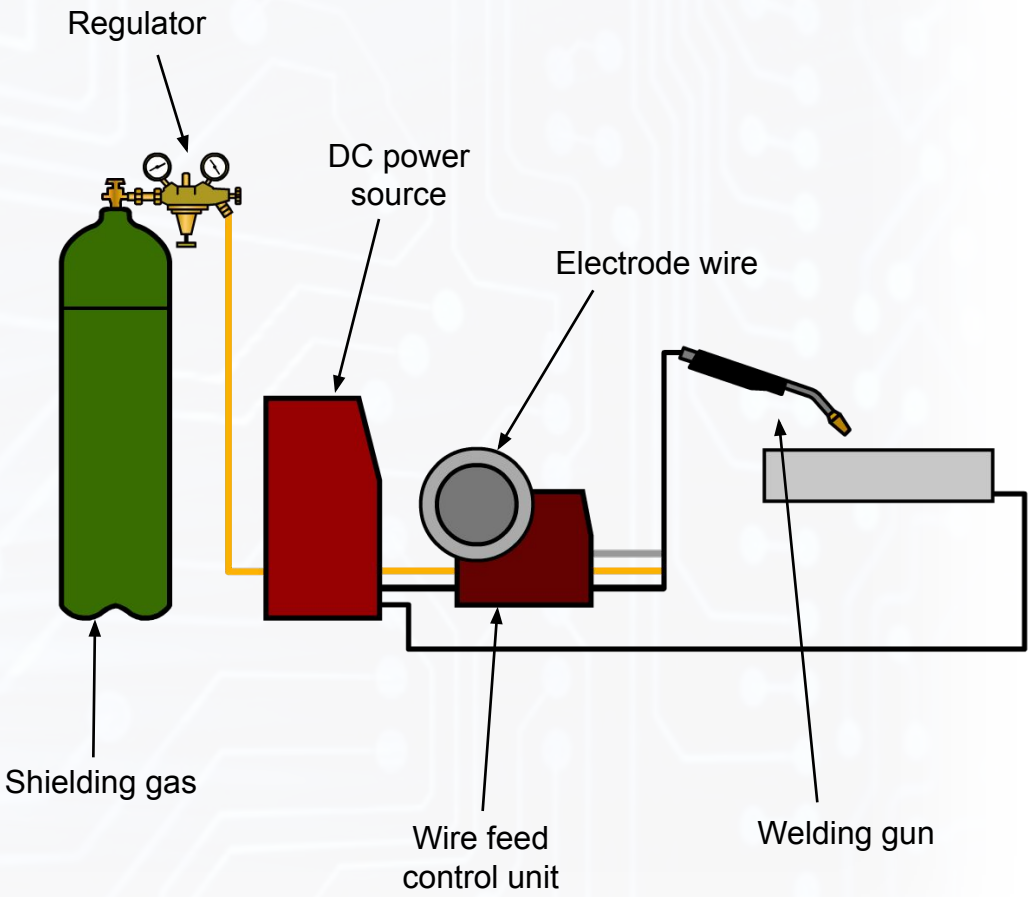


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MIG Welding Equipment

These components show a typical MIG welding set-up:

- *Constant voltage DC power source*
- *Shielding gas cylinder and regulator*
- *Electrode wire*
- *Wire feed spool and control unit*
- *Welding gun*



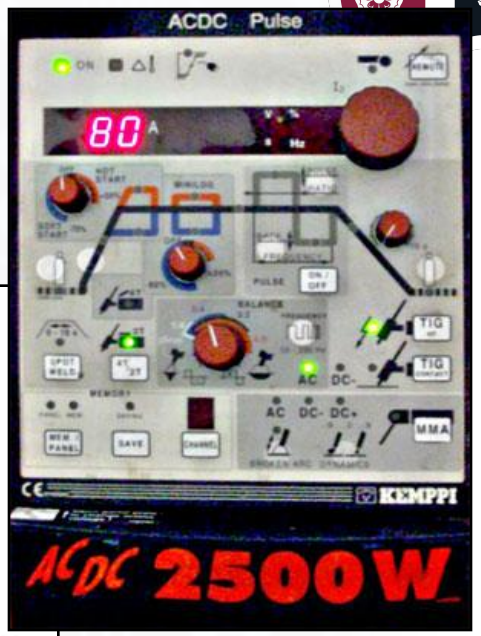
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Setting up the Equipment

The type of welding application determines several things:

- *The welding voltage*
- *The speed of the electrode wire*
- *The flow rate of the shielding gas*







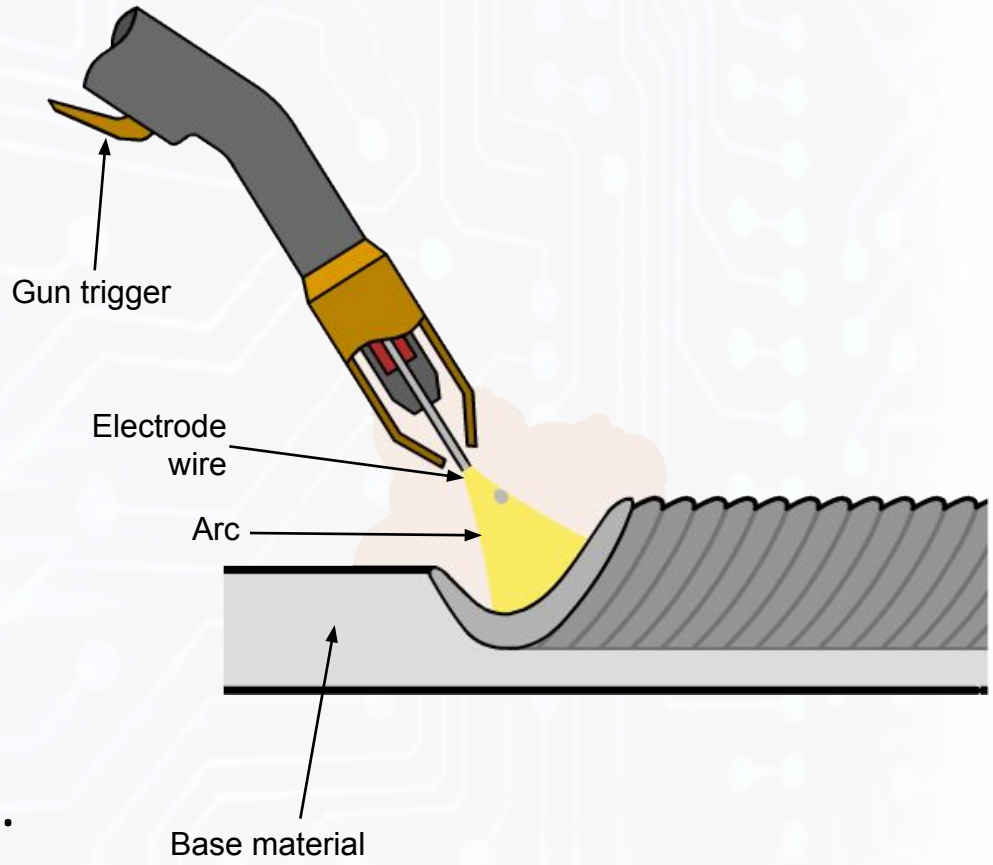
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The MIG Welding Process

The welding gun trigger controls the electric arc, the wire feed and the gas flow.

The trigger is squeezed to strike an electric arc between the base material and the electrode wire.

Heat from the arc melts the electrode wire and the surface of the base materials.



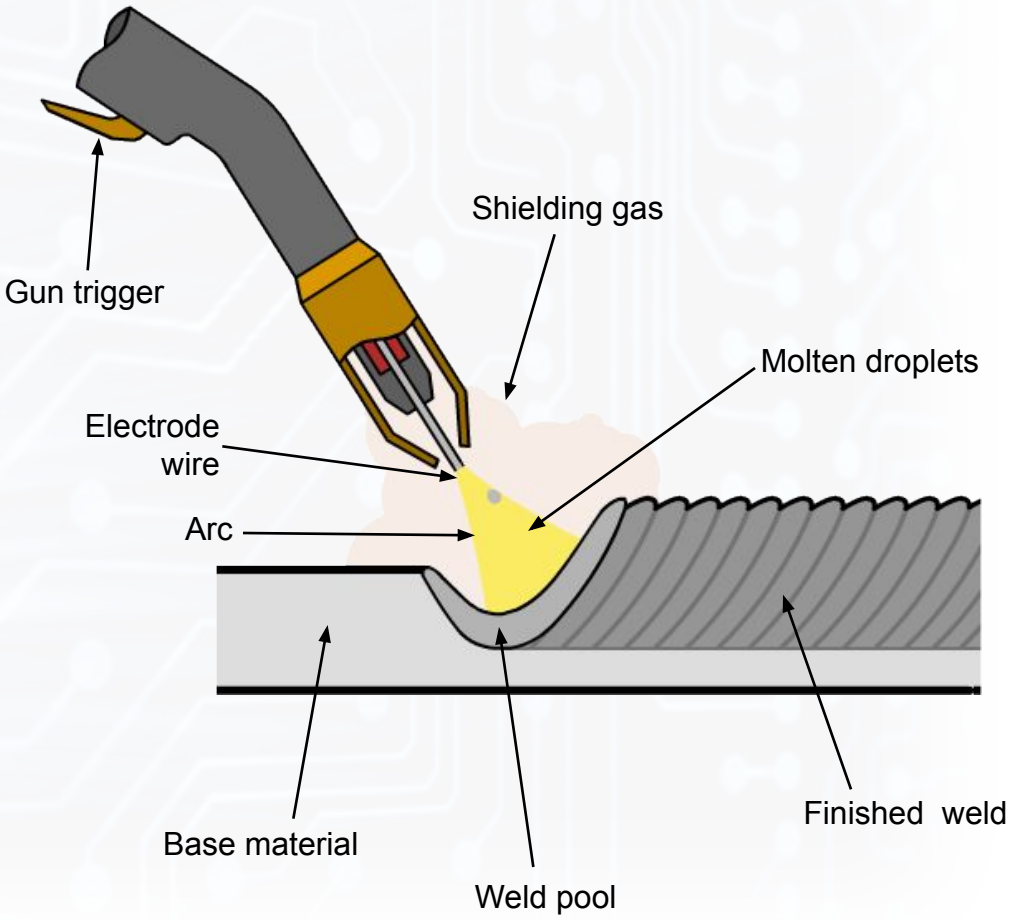
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The MIG Welding Process

Molten droplets from the electrode wire are transferred to the weld pool.

The shielding gas protects the weld pool from air to produce a weld without slag.

The gun trigger is released when the weld is completed.



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Question 1

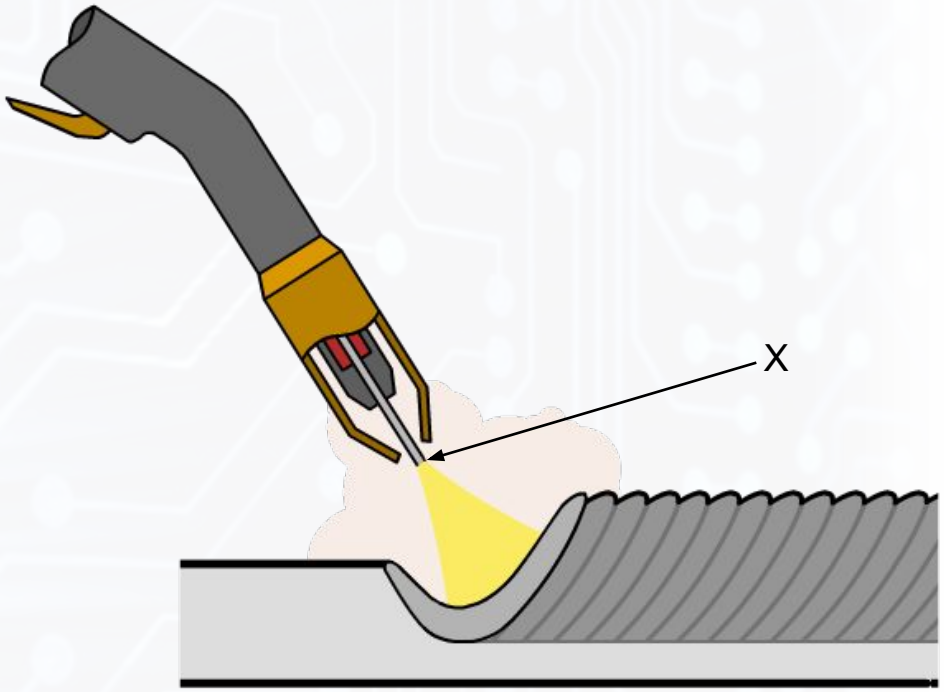
The diagram shows the MIG welding process. What is X?

A) The weld pool

B) The electrode wire

C) The shielding gas

D) The electric arc



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Question 1

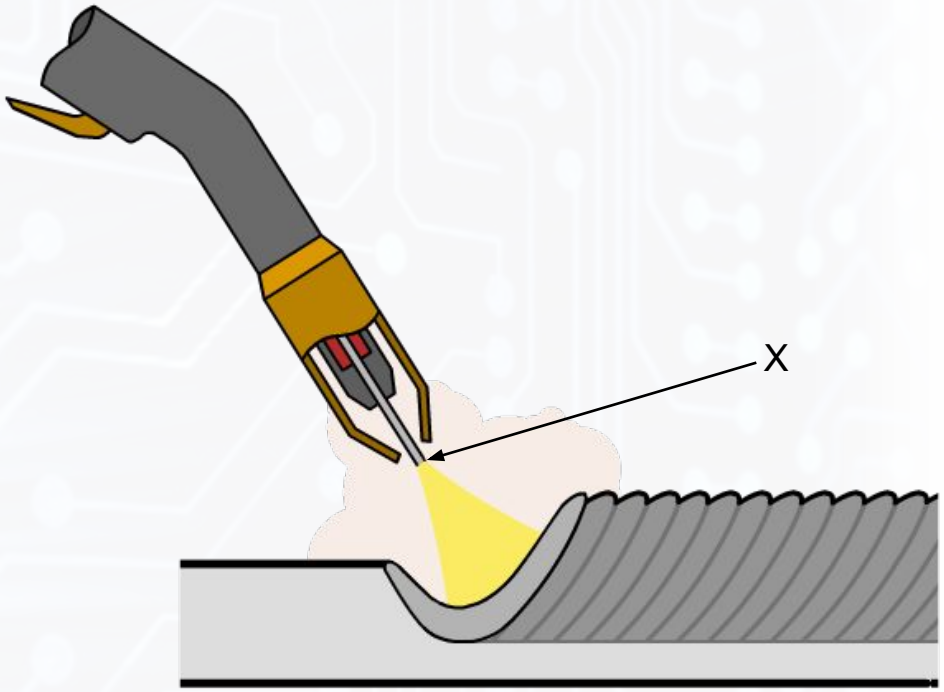
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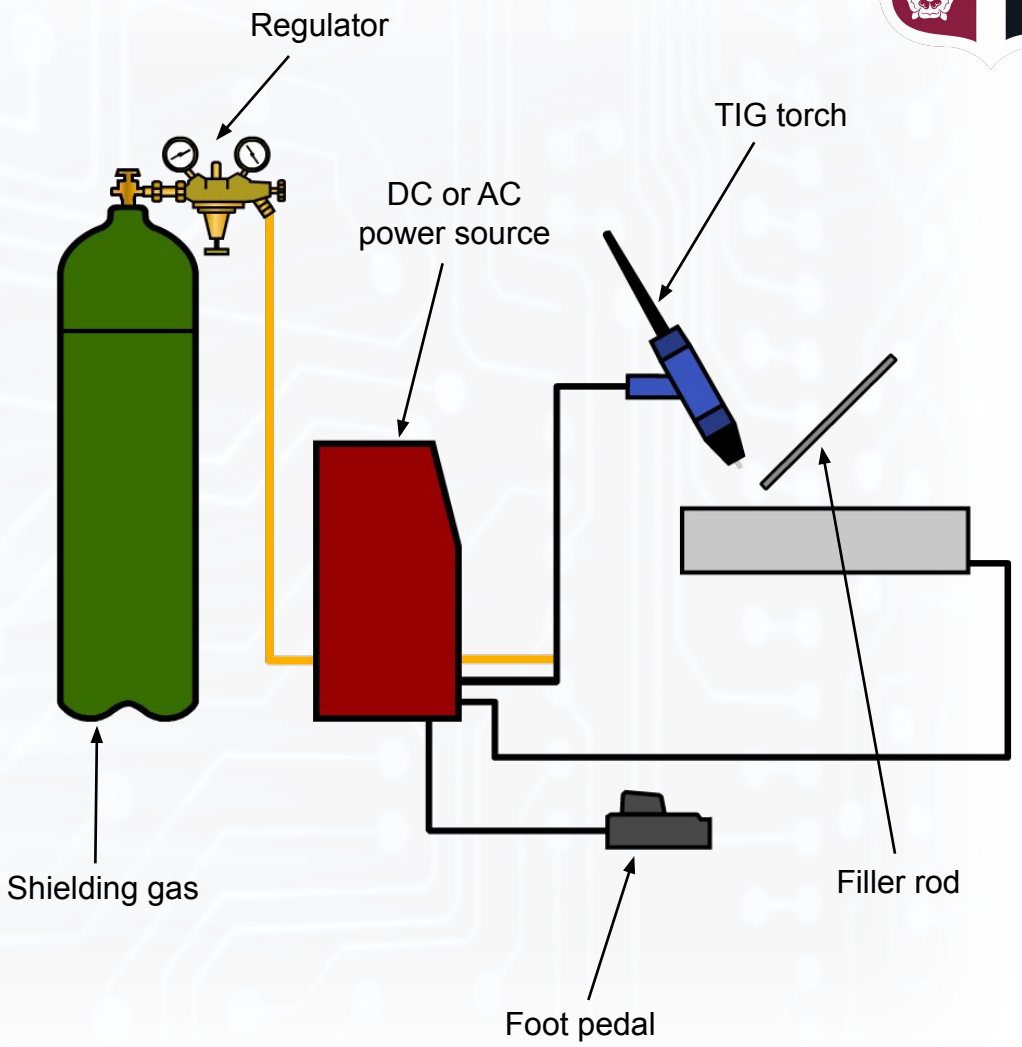


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TIG Welding Equipment

These components show a typical TIG welding set-up:

- *Constant current AC or DC power source*
- *Shielding gas cylinder and regulator*
- *TIG torch with tungsten electrode*
- *Foot pedal*
- *Optional filler rod*



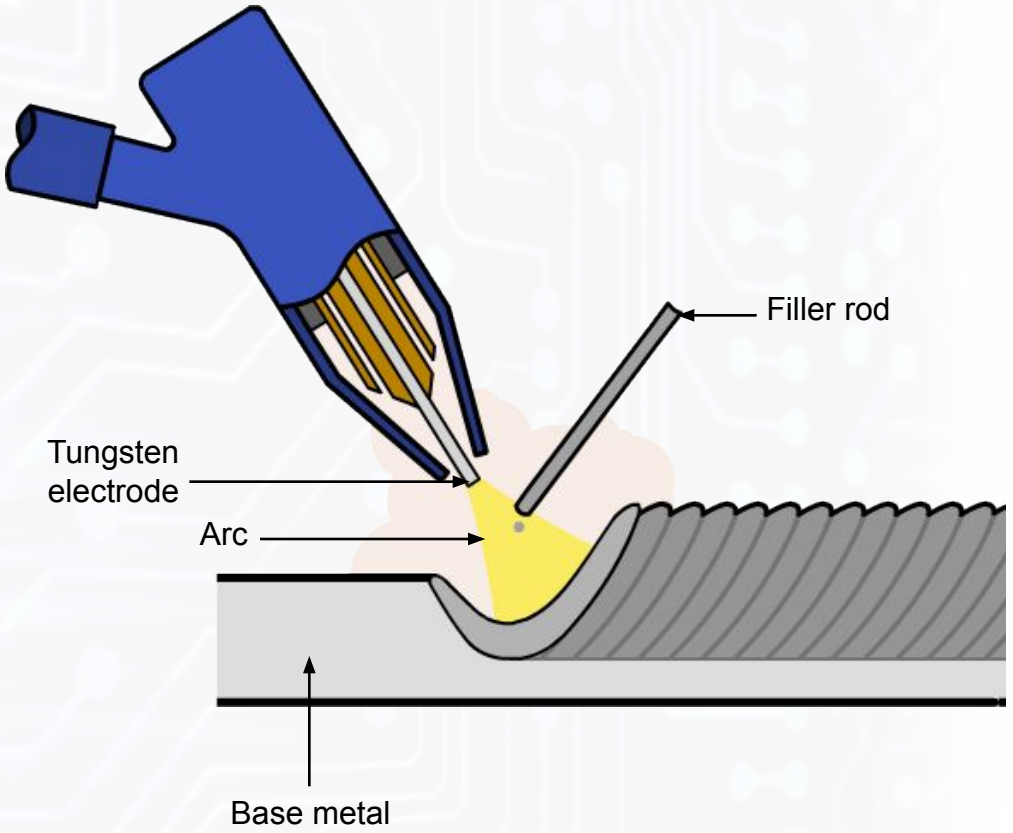
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The TIG Welding Process

The TIG torch is held in position ready to weld.

When the foot pedal is pressed, it turns on the shielding gas and strikes an electric arc between the base material and the tungsten electrode.

Sometimes, a filler rod may be used to add extra material to the weld.



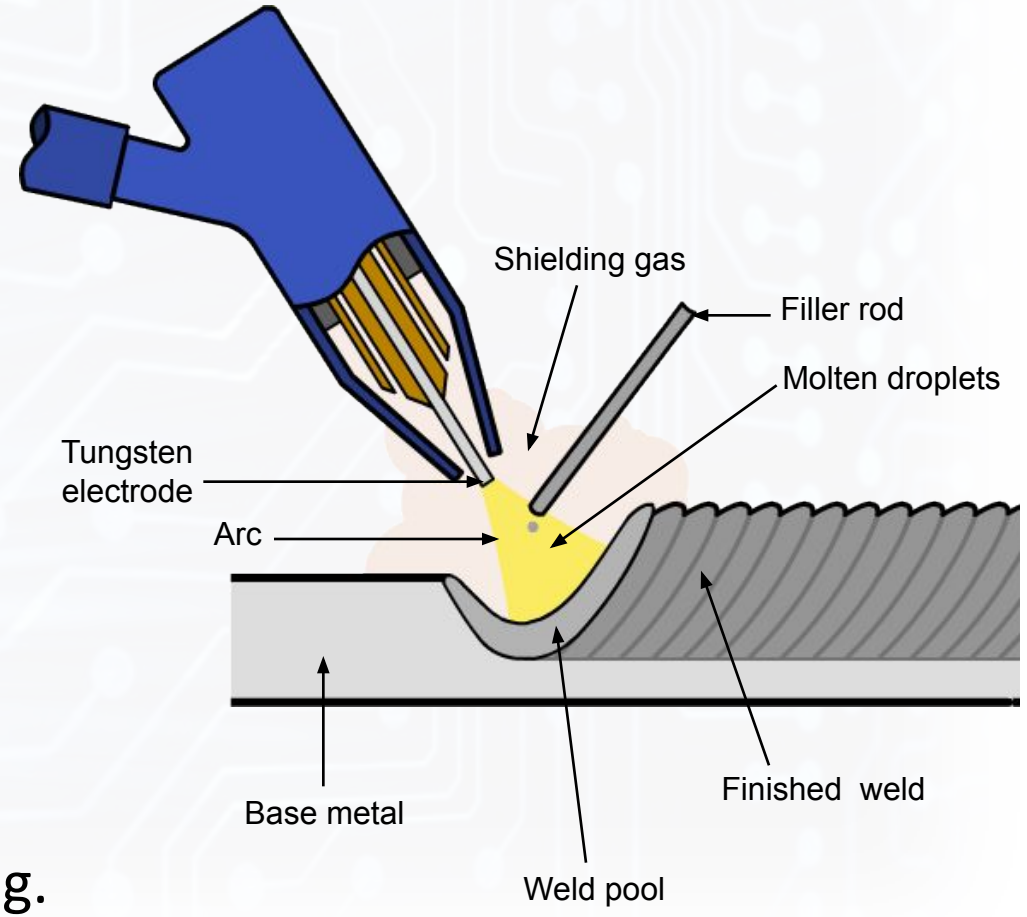
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The TIG Welding Process

Heat from the arc melts the surface of the base material to create a weld pool.

Arc heat also melts the filler rod (if used), and molten droplets are transferred to the weld pool.

The shielding gas protects the weld pool from air to produce a weld without slag.



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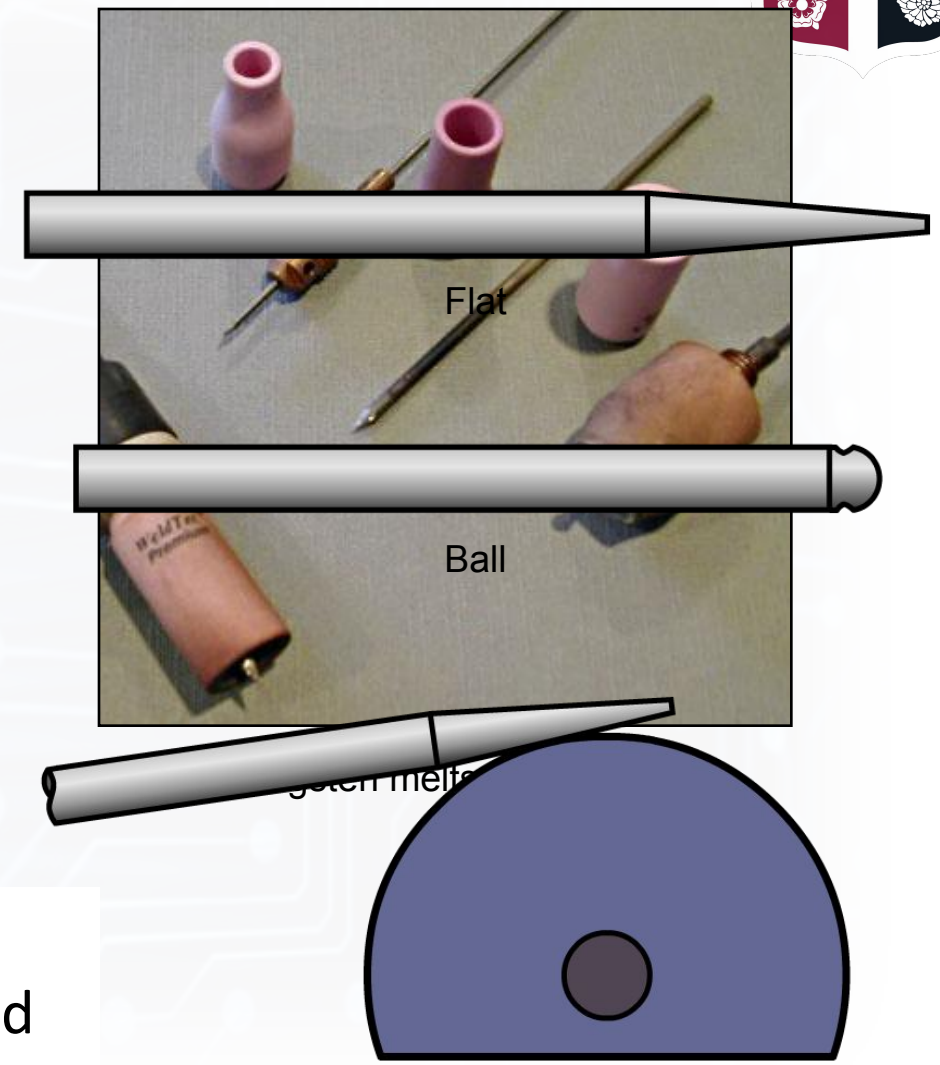
TIG Electrodes

TIG electrodes are made from tungsten or a tungsten alloy.

The choice of electrode depends on the type and thickness of the base material and the type of power source.

The shape of the electrode tip determines weld width, penetration and quality.

The welding process causes material to be added to or eroded from the tip.



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Question 2

During the TIG welding process, where would an electric arc be created?

A) Between the base material and a filler rod

B) Between the base material and a gas nozzle

C) Between a tungsten electrode and a filler rod

D) Between the base material and a tungsten electrode

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Question 2

During the TIG welding process, where would an electric arc be created?

A) Between the base material and a filler rod

B) Between the base material and a gas nozzle

C) Between a tungsten electrode and a filler rod

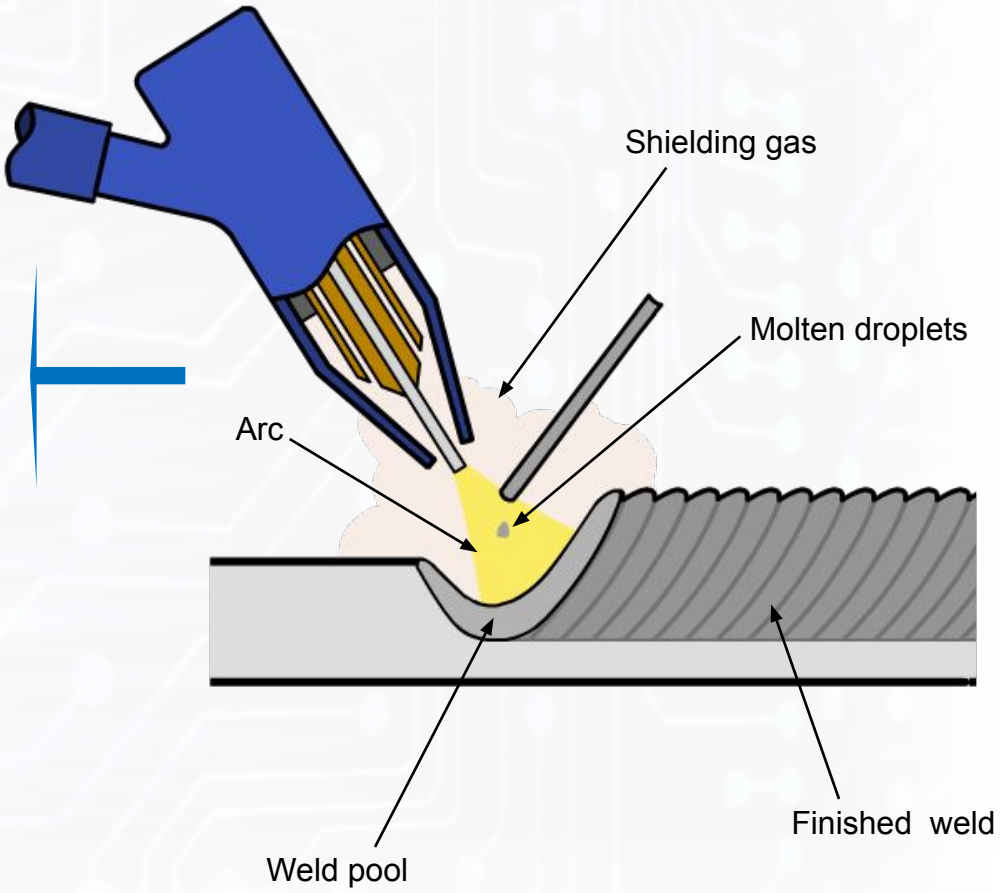


D) Between the base material and a tungsten electrode

Shielding Gas Function

The shielding gas has several functions:

- *To protect the weld pool from air*
- *To stabilise the electric arc*
- *To determine the metal transfer type*
- *To affect the quality of the weld finish*
- *To affect the travel speed in producing the weld*



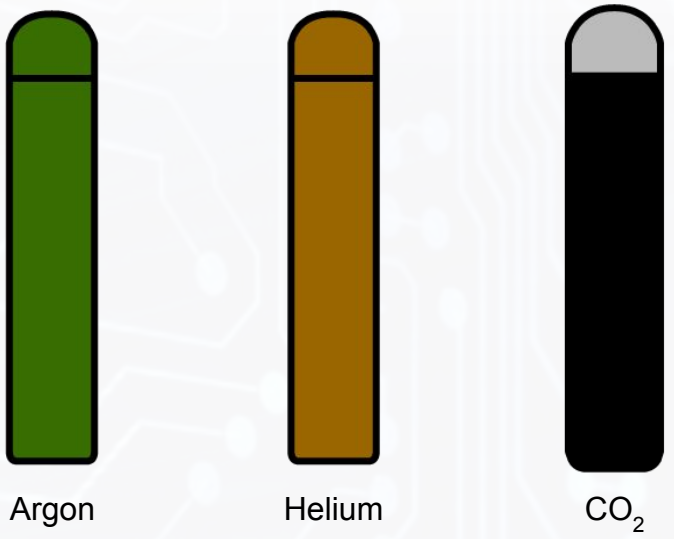
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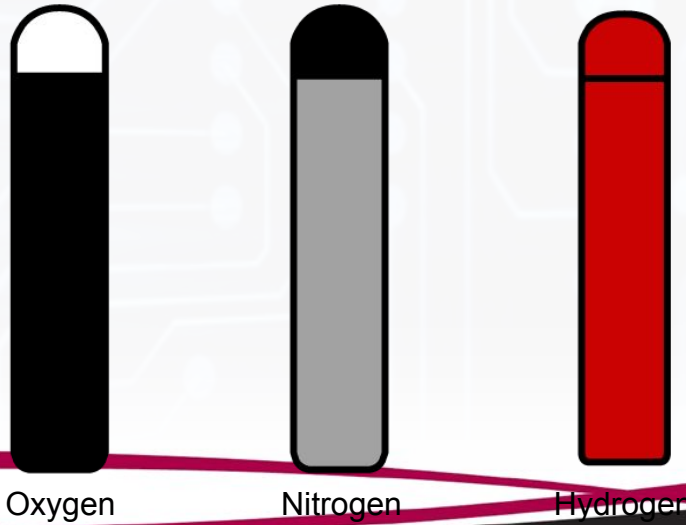
Types of Shielding Gas

Choosing the correct shielding gas is essential for successful welding.

The three gases used for shielding are argon, helium and carbon dioxide.

Other gases, such as oxygen, nitrogen and hydrogen, may be added to modify arc, weld and metal transfer characteristics.





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MIG/MAG Shielding Gases

The shielding gas used in MIG welding is usually argon or an argon-helium mixture.

If an active gas such as oxygen or carbon dioxide is mixed in, the term used to describe the welding process is MAG (metal active gas) welding.

Argon in %	Helium in %	Carbon dioxide in %	Oxygen in %	Material in mm
Mild steel				
93		5	2	≤ 4
86		12	2	4 – 10
78		20	2	≥ 10
Aluminium				
70	30			≤ 6
50	50			6 – 10
25	75			≥ 10
Stainless steel				
13.5	85	1.5		≤ 3
43	55	2		3 – 10
60	38	2		≥ 10

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TIG Shielding Gases

The shielding gases used in TIG welding must be inert.

Active gases cause the tungsten electrode to oxidise.

Pure argon or helium, or a mixture of both, may be used for steel and aluminium welding.

Small amounts of nitrogen or hydrogen may be mixed in to improve weld characteristics.

Argon in %	Helium in %	Hydrogen in %	Material
Mild steel			
100			Thin
75	25		Thick
100			General
Aluminium			
100			Thin
75	25		Thick
100			General
Stainless steel			
95		5	Thin
75	25		Thick
95		5	General

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Question 3

Which combination of gases is best for TIG welding thick mild steel plates?

A) 85% argon and 5% hydrogen

B) 75% argon and 25% carbon dioxide

C) 75% argon and 25% helium

D) 25% argon and 75% helium

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Question 3

Which combination of gases is best for TIG welding thick mild steel plates?

A) 85% argon and 5% hydrogen

B) 75% argon and 25% carbon dioxide



C) 75% argon and 25% helium

D) 25% argon and 75% helium

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Advantages of TIG Welding





The TIG welding process produces high quality welds that can be made without using a filler material.

TIG welding is a fairly clean process producing minimal sparks, smoke and fumes.

TIG welding is ideal for thin metals and is quite a reliable process.

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Disadvantages of TIG Welding





TIG welding is complex to set up, requires high skill levels and is expensive.

TIG welding is a slow process and not ideal for thick metals.

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Summary

In this presentation you have seen:

- *the MIG welding process*
- *the TIG welding process*
- *the shielding gases used in MIG and TIG welding*

End